

Bappy Chandra Debnath

Kaderabad Housing, Mohammadpur, Dhaka 1207

Phone: 01797127296 — **Email:** bappynath2001@gmail.com

GitHub: github.com/bappyBDN — **Portfolio:** bappybdn.github.io/portfolio

LinkedIn: linkedin.com/in/bappy-chandra-debnath-3263063b9



EDUCATION

Ahsanullah University of Science and Technology

Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering (CGPA: 3.25/4.00)

Expected June 2026

Relevant Coursework: AI Engineering, LLM Fine-tuning, AI Agents, Software Engineering Principles, Full-stack Development

PROJECTS & EXPERIENCE

OmniCart AI — Multi-Agent Swarm for E-commerce

Personal Project

- Architected a LangGraph-powered multi-agent system managing autonomous customer query routing, live SQLite inventory tracking, and vector-based policy retrieval (ChromaDB RAG).
- Integrated Llama-3.2-Vision for instant product image identification and structured context extraction.
- Crafted an interactive Streamlit Admin Dashboard featuring real-time chat history sync and a "Human-in-the-Loop" (HITL) take-over trigger mechanism.

AgroPulse-BD — Deep Learning Remote Sensing

Lead Developer / Researcher

- Engineered a predictive decision support system utilizing Bi-LSTM and 1D-CNN architectures for high-precision flood forecasting and crop health monitoring in the Sunamganj region.
- Synthesized 15 years of multi-modal data, integrating Sentinel-1 SAR and Sentinel-2 NDVI datasets via daily scheduled GitHub Actions pipelines.
- Optimized deep learning models using Focal Loss and implemented model explainability through SHAP; deployed Google Gemini via API for localized Bengali warnings.

AI-Powered Gmail Agent — RAG System

Project Lead / Developer

- Spearheaded a RAG-driven AI assistant using Groq (Llama 3) and Sentence Transformers to generate professional replies from a Supabase vector database.
- Implemented a robust Human-in-the-Loop workflow with a 1-5 star operator rating feedback loop to refine future prompts dynamically.

AI-Powered Career Information Bot — Workflow Automation

n8n Automation

- Deployed an end-to-end automation pipeline using n8n to capture and extract career links from Telegram channels using the Llama-3.1-70B model via Groq APIs.
- Streamlined structured data ingestion into Google Sheets and configured memory tracking the last 30 user interactions.

Home Chef Hubs — Full-Stack Web Development

Academic Project

- Designed a full-stack web platform using .NET and C# designed to securely bridge home chefs with local customer networks.
- Executed a scalable relational backend and database schema to manage complex user sessions and customer relationship handling.

LEADERSHIP & ACTIVITIES

Ahsanullah University of Science & Technology

Mar 2022 – Present

- Led cross-functional development teams for major projects, including AgroPulse-BD, coordinating model implementation and deployment.
- Directed full-stack development efforts, backend architecture planning, and collaborate with faculty mentors on multi-modal data fusion research.

SKILLS & INTERESTS

Programming & Tools: Python, C++, C, .NET, C#, PHP, SQL, n8n, FastAPI, Docker, PySpark, Git Hygiene, GitHub Actions

AI/ML & Architecture: LangGraph, LangChain, Deep Learning (Bi-LSTM, 1D-CNN), RAG, ChromaDB, Supabase, LLM Fine-tuning

Problem Solving: Solved 100+ problems across LeetCode, NeetCode, Codeforces, & AtCoder

Frontend & UI/UX: Streamlit, Modern HTML, CSS, UI/UX Design

Interests: Agentic AI, Geospatial Analytics, Human-AI Collaboration